

MODEL TRAINING REPORT

Electronic Nose Coffee Classification System

Model: Support Vector Machine (SVM)

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TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Time
96.9%	91.7%	0.0s

MODEL CONFIGURATION

Model Type	Support Vector Machine (SVM)
Epochs	200
Support Vectors	18
Final Training Loss	0.07952
Data Split	80% Training / 20% Validation

GAS SENSORS USED (8 Sensors)

TGS2600	MQ135	MQ3	MQ6
MQ7	TGS2602	TGS2611	TGS2620

MODEL EVALUATION RESULTS

Training Data — Accuracy: 96.88%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9592	0.9787
Recall	0.9792	0.9583
F1-Score	0.9691	0.9684
Support (n samples)	96	96

Validation Data — Accuracy: 91.67%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.8846	0.9545
Recall	0.9583	0.8750
F1-Score	0.9200	0.9130
Support (n samples)	24	24

Confusion Matrix (Validation Data)

	Predicted: High	Predicted: Low
Actual: High	23	1
Actual: Low	3	21

TP = True Positive (High predicted as High)

FP = False Positive (Low predicted as High)

TN = True Negative (Low predicted as Low)

FN = False Negative (High predicted as Low)

MODEL DETAILS & TRAINING CURVES

SVM Summary -- 200 Epochs

Support Vectors	18
Final Training Loss	0.079519
Total Epochs	200

Feature Weights (Weight Magnitudes -- Top 15)

No	Feature Name	Weight (proportional bar)
1	tgs2600-std	0.3454
2	tgs2602-std	0.2976
3	tgs2600-mean	0.2787
4	tgs2620-std	0.2618
5	tgs2600-max	0.2373
6	tgs2600-range	0.2242
7	mq6-std	0.2074
8	mq7-max	0.2058
9	tgs2600-median	0.2001
10	mq7-mean	0.1741
11	tgs2620-median	0.1674
12	tgs2600-min	0.1604
13	tgs2602-range	0.1592
14	tgs2611-std	0.1549
15	mq7-median	0.1542

Accuracy Curve (Checkpoint)

Epoch	Training Accuracy (%)	Validation Accuracy (%)
20	96.88	91.67
40	96.35	91.67
60	96.35	91.67
80	96.88	91.67
100	96.88	91.67
120	96.88	91.67
140	96.88	91.67
160	96.88	91.67
180	96.88	91.67
200	96.88	91.67